

Federative Coordination and Inequality of Provision: Brazil's Education Policy Response to the COVID-19 Pandemic

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The COVID-19 pandemic posed major challenges to federal systems, especially in balancing central and subnational responsibilities in education. This study introduces the Remote Education Program Index (REP Index), a novel metric to assess the quality and comprehensiveness of remote education policies adopted during the pandemic. Drawing on the Brazilian case, we evaluate daily data from all 27 subnational units between March 2020 and December 2021, and show that the absence of federal coordination and the loosening of fiscal obligations led to reductions in education spending and contributed to stark disparities in the quality of remote education programs. Our findings underscore that decentralized education management, in the absence of national guidance and redistribution, can exacerbate existing territorial inequalities during emergencies. The REP Index provides a valuable tool for evaluating remote learning policies and offers broader lessons for federations seeking to build more equitable education systems in times of crisis.

The COVID-19 pandemic required an immediate response from governments worldwide. Governments hurriedly implemented a diverse and comprehensive set of policies to impede the spread of the virus and mitigate its effects on the

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Received June 26, 2023; revised October 8, 2025; accepted October 16, 2025; electronically published February 13, 2026

Comparative Education Review, volume 70, number 1, February 2026.

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population and economic activity (Capano et al. 2020). As the health emergency crisis persisted, federal republics faced additional challenges in balancing the decision-making process between central and subnational governments. Whereas excessively centralized decisions may lead to policies unsuited to the heterogeneous epidemiological reality of each subnational unit (Choutagunta et al. 2021), a lack of federal coordination can also hinder the capacity of states and municipalities, particularly poorer states and municipalities, to implement necessary interventions. It is crucial for federations to carefully consider the trade-offs from decentralization in policy making during prolonged emergencies.

One of the main policies adopted worldwide to contain the rising infections and deaths caused by the COVID-19 pandemic was the closure of on-site schools and a transition to off-site schooling. Although this policy effectively reduces virus transmission by reducing in-person interactions (Gurdasani et al. 2021), institutions providing education services had to implement off-site teaching and learning programs quickly (World Bank 2020). Even in contexts where distance education reached most students, it exacerbated existing inequalities due to barriers in access to technology in poor environments (Li et al. 2023). Some studies analyzing the response of federal republics to the COVID-19 pandemic have argued that one of the advantages of federalism was that it enabled subnational units to adopt policies better suited to local needs and to deploy programs more quickly (Bennouna et al. 2021; Angelici et al. 2023; Lachance 2024). However, researchers have argued that national guidelines and cooperation are necessary to ensure minimum standards are met during emergencies (Zheteyeva et al. 2017; Benton 2020; Bowling et al. 2020; Greer et al. 2021). When the federal government does not mandate minimum standards, education provision gaps can widen between subnational units, especially in countries with marked inequalities, as recent studies have shown in the case of COVID-19 (Bonaf and González 2020; Acuña Ortigoza 2021; Zancajo et al. 2022; Ranieri 2024).

Although there have been considerable efforts to analyze the impact of COVID-19 on learning outcomes, there have been limited efforts to measure and evaluate remote education programs to identify how these programs' design and implementation contributed to inequalities across various contexts and the prolonged COVID-19 emergency (Zancajo et al. 2022). As we show in this study, off-site education is distinct from on-site schooling, and evaluations of education federalism during COVID-19 need to consider the specific impact of these programs separately from public education systems

The authors would like to thank Ingrid Castro and Dara Aparecida Vilela Pinto for valuable discussions and research assistance on this project, and the *Rede de Pesquisa Solidária em Políticas Públicas e Sociedade*. This work was supported by the FAPESP, grant numbers 2013/07616-7, 2021/08772-9, and 2022/06523-4. The funding received by the researchers did not directly contribute to the study design, data collection, analysis, data interpretation, manuscript writing, or the decision to submit the manuscript for publication. The authors declare no competing financial interests.

that rely on traditional on-site education provision. To contribute, we propose a new measure of the quality and comprehensiveness of remote education programs. Using data from state remote education programs adopted during the COVID-19 pandemic in Brazil, we show that the lack of central government coordination drives remote education quality in the context of federalism.

Our study is based on indicators measuring the timing, duration, and types of government remote schooling programs implemented in state public schools during the COVID-19 pandemic in Brazil (Barberia, de Paula Moreira et al. 2021). Specifically, we measure state-level daily scores for remote education programs (including broadcast means, access, and coverage), which are part of the indicators reported in the COVID-19 Government Response Tracker for the Brazilian Federation (CGRT-BRFED), starting on the date of school closures in 2020 and the beginning of the school year in 2021, and extending until the end of the school year in both years. We calculate the mean for each period, thus obtaining a measure of both quality, early response, and the reach of the measures adopted. We examine whether investments were made to provide access (such as the provision of phones, mobile chips, tablets, books to students and teachers, and internet subsidies), whether policies were enacted to guarantee student supervision (by teachers or by education departments), and the scope of coverage of these programs (such as kindergarten, elementary, and high school education). We then use these indicators to create a Remote Education Program Index (REP Index) that captures the multiple dimensions of remote education programs' overall quality. We record the programs of the 26 states of the Brazilian Federation and the federal district from March 1, 2020, to December 31, 2021.

Based on this data, our study makes three significant contributions. First, we demonstrate that the absence of federal coordination has led to state programs of remote education of poor quality that are regionally unequal. Second, our findings demonstrate that there was a net reduction in state spending on education because the Brazilian federal government modified the minimum mandatory education expenditures during COVID-19. Finally, we show that, in the context of a lack of federal coordination on technical decisions and the loosening of constitutional spending obligations, states adopted remote education programs that reflected their preexisting capacity levels, exacerbating the already significant regional inequalities in education provision.

The rest of this article is organized as follows. The next section discusses the literature on federalism and the resilience of national and subnational governments to respond to emergencies concerning schooling provision. We then introduce our theoretical expectations for the case of Brazil, where the federal government adopted a *laissez-faire* approach to remote learning. The data and models used to test our hypotheses are then presented, and the results are discussed in the following section. The conclusion identifies future directions for research and policy evaluations.

Literature Review

The Response of Federations to Emergencies: The Case of the COVID-19 Pandemic

Federal systems inherently face tensions when responding to national emergencies, particularly in sectors like education. Crises further complicate the delicate balance between subnational autonomy and national coordination. Comparative research during normalcy periods shows that federation education governance struggles to reconcile local political autonomy with the need for centralized direction to ensure equitable outcomes across regions (Tilak 1989). The COVID-19 pandemic intensified these dilemmas, as governments fought over how to intervene in domains typically managed by states and municipalities.

Scholars have argued that decision making becomes more centralized during intense socioeconomic upheaval in federative countries ('t Hart et al. 1993; Hegele and Schnabel 2021). Greer et al. (2021) show that, during the first wave of COVID-19, several countries centralized decisions on prevention of transmission and governance, with few moving in the opposite direction. Centralization was said to help contain transmission across a federation and ensure assistance reached areas most in need.

There is an important debate on the positive and negative effects of decentralization/centralization of public policy in federal countries. Decentralization is argued to make policies better suited to local needs but may also increase interregional or local inequalities (Peterson 1995; Weingast 1995). The inequality or equity of income among federal entities depends on the model of fiscal federalism. Federations differ in which governments can collect taxes, how revenues are distributed, and the existence of equalizing central funds to minimize inequalities (Hinchliffe 1989). Whether federalism follows a dual, cooperative, or competitive model (Deeg and Lütz 2000; de Gadelha 2018), the central government is critical in ensuring equity across heterogeneous regions. Federal countries can reduce inequalities through centralized fiscal systems (Hinchliffe 1989; Treisman 2007). In other words, contrary to what some authors have argued, federalism does not necessarily imply additional disparities (Wildavsky 1984). Institutional arrangements and intentional leveling policy making can produce the opposite effect.

In the area of education, scholars have increasingly been pointing to the need to deepen discussions on the effects of decentralization beyond the differences between federal and unitary states (Carnoy et al. 2017; Bray 2023). Federations tend to have decentralized administration and financing of education (Hinchliffe 1989; Winkler 1989), but there are significant variations in the role of central governments in policy design and monitoring, as well as the degree to which fiscal centralization is used to reduce regional inequalities (Prawda 1993).

Before COVID-19, research on emergency responses and education provisions was limited (Gall et al. 2022). In high-income federations like the United

States, Zheteyeva et al. (2017) note that few schools or districts had procedures to ensure educational continuity during emergencies. Most systems lacked policies for partial openings, alternate facilities, and access to distance education—including devices, internet, temporary transfers, and services like school meal replacements. These weaknesses underscore the important role national governments can play in coordinating and ensuring an effective educational emergency response.

In low- and middle-income countries, where education systems often rely on subnational implementation, coordination gaps are exacerbated by limited state capacity. Ganimian et al. (2024) show that even modest interventions to strengthen frontline administrative structures at the local level can significantly improve early childhood education services. These findings highlight the critical role of subnational institutional capacity—not only in routine operations but especially during systemic shocks. In federal systems, variation in state capacity becomes a key factor shaping the equity and effectiveness of emergency responses. Furthermore, research comparing e-learning policies in the United States and the European Union shows that structural governance differences profoundly influence how educational technologies are integrated into national strategies (Roumell Erichsen and Salajan 2014).

Studies show that, across different national contexts, the COVID-19 pandemic created an unexpected shock, forcing education actors to make rapid decisions under intense pressure (Kim et al. 2021). In response to this reality, scholars have evaluated the pandemic's impact on education. Zancajo et al. (2022) analyzed long-term education policy responses to the COVID-19 crisis in Europe and identified three main areas of intervention: digitalization, combating inequalities, and teacher development. In Chile, closures of early childhood centers led to measurable developmental losses in language and executive function among vulnerable children, showing the pandemic can have long-term effects on education inequality (Abufhele et al. 2024). Studies indicate that effective intergovernmental coordination was better for public policy responses (Greer et al. 2023). In contrast, federal governments that adopted adversarial stances—marked by competition with subnational governments—saw weaker policy outcomes (Benton 2020; Bowling et al. 2020). Research in low-income federal countries confirms that preexisting learning inequalities were widened (Ranieri 2024). The available evidence highlights the importance of understanding which types of educational policy designs were adopted in federative contexts.

The Case of Brazil

The Brazilian experience during the COVID-19 pandemic offers a relevant case for analysis due to the country's internal heterogeneity. In federations like Brazil, where education policy and financing are decentralized, states and municipalities play central roles in shaping outcomes. As Carnoy

et al. (2017) argue, subnational comparisons within federal systems are essential for revealing meaningful variations in policy implementation, administrative capacity, and educational equity. Harman's (1985) analysis of Australia and the United States shows how structural differences within federations shape policy outcomes, highlighting the value of subnational comparative research for understanding decentralized education systems. In addition, Asim et al. (2023) emphasize the importance of examining the "missing middle" in education systems, underscoring the need to understand the functions and limits of intermediate institutional actors. National averages may obscure intranational disparities; thus, analyzing Brazil's internal dynamics enables a more accurate assessment of the pandemic's impact and generates insights with broader comparative relevance.

The 1988 constitution defined the responsibilities of the national government and the federative entities in formulating and executing education policies, characterized by a mix of centralization and decentralization in administration, financing, and implementation (de Almeida 2005, Arretche 2009). Article 211 of the constitution and Amendment 14 of 1996 establish that states are exclusively responsible for secondary education but share responsibility with municipalities for primary education. Municipalities are solely responsible for early childhood education (kindergarten). States and municipalities handle spending and providing elementary education. They also organize instruction (curricula, textbooks, teaching methods, schedule), manage personnel (hiring, pay scales, assigning teaching responsibilities, training), and oversee school planning and administration. The Ministry of Education oversees the entire primary and secondary system, including establishing minimum standards, approving the national curriculum, and administering evaluations.

To finance the education sector, the central government acts as formulator, coordinator, and supervisor, controlling policy implementation through revenue transfers and policy design, whereas states and municipalities play more administrative roles with limited fiscal autonomy (Machado and Palotti 2015). In a system considered fiscally centralized (Soares and Machado 2020), the central government controls most revenue collection and holds the power to transfer funds to poorer states (Arretche 2009). States and municipalities must allocate at least 25 percent of their taxes to education, and the central government must allocate 18 percent of its revenue. These constitutional requirements establish a minimum spending floor and help reduce disparities among local governments. Since Amendment 14 of 1996, measures were taken to introduce redistributive mechanisms to the national education network and promote equal educational opportunities and a minimum quality standard through the Education Ministry's technical and financial assistance to states, the federal district, and municipalities. Interstate redistribution and constitutional spending obligations have reduced the inequality in per capita spending and municipal outcomes in education (Arretche 2016).

In contrast to research that argues that decision making becomes more centralized during a crisis, the government of Jair Bolsonaro did not seize the extraordinary event of the COVID-19 pandemic as an opportunity for further centralization within the Brazilian Federation (Grin et al. 2022). During the early response to COVID-19 in 2020, and as the crisis continued in 2021, the federal government adopted a *laissez-faire* approach to education policy—defined here as the combination of a lack of national coordination in technical decisions and the loosening of constitutional obligations on education spending. Rather than asserting leadership or issuing minimum standards, the federal government refrained from guiding subnational responses and allowed states considerable discretion in how to manage the crisis.

This dynamic must be understood considering Brazil's federal institutional structure. Although the Constitution guarantees political and administrative autonomy to subnational entities and prevents hierarchical federal intervention, coordination remains possible through policy guidance and intergovernmental mechanisms. In the health sector, the Unified Health System ensures structured spaces for federal coordination and financial induction. By contrast, the education sector lacks a national system or institutionalized arenas for vertical collaboration. Although the Ministry of Education holds regulatory authority and can offer technical and financial support, these tools were not meaningfully deployed during the pandemic, reinforcing fragmented subnational responses.

In education, the 27 subnational units (26 states and the federal district) mandated on-site school closures, followed by the introduction of remote education. Despite the federal government's *laissez-faire* approach to managing the COVID-19 emergency, it allocated significant resources to subnational governments. Congress passed Law 173/2020 in May 2020, transferring R\$60.15 billion (approximately US\$10.29 billion) to states. The law suspended debt payments to the central government and maintained participation fund transfers (Fundo de Participação dos Estados / State Participation Fund and Fundo de Participação dos Municípios / Municipal Participation Fund, respectively) at prepandemic levels, totaling R\$16 billion (approximately US\$2.7 billion). Therefore, despite the significant loosening of education policy obligations, the fiscal institutional regime of cooperation showed resilience.

Although decentralized systems provide local autonomy, they risk fragmentation and uneven policy responses when national coordination is weak or absent. In education, balancing autonomy with national standards is especially sensitive, as local actors deliver services, but systemic inequalities often require coordinated interventions. During emergencies like the COVID-19 pandemic, the ability—or failure—of central governments to organize coherent responses can deeply shape educational outcomes across regions.

Building on this framework, our study investigates how Brazil's federal structure influenced public education management during the COVID-19

pandemic. Specifically, we ask: How did the absence of federal coordination affect the implementation and quality of remote education programs in Brazil during the pandemic?

Hypotheses

As emphasized in the preceding sections, two distinct expectations exist regarding how decentralization influences state-level education policies. On the one hand, the increase of fiscal and administrative powers to states may enable responses better tailored to local needs, leading to improved policies (Weingast 1995). Conversely, the lack of central government guidance and coordination may leave states, especially those with lower capacities, unprepared to manage emergency education provision, leading to significant volatility among subnational responses (Hinchliffe 1989; Treisman 2007). During the COVID-19 pandemic, the federal government removed minimum spending requirements and did not implement a national program or issue guidelines on the minimum standards for off-site education. As a result, states could reduce education expenditures, including remote education programs. Alternatively, they could maintain or expand allocations, because the federal government supplemented state budgets to offset tax revenue losses. Therefore, we first test the following hypothesis:

- HYPOTHESIS 1A. The laissez-faire management of the COVID-19 pandemic in education by the central government will lead to state-level cuts in education expenditures.

Because on-site schooling was not provided to students in 2020 and only partially in 2021, states developed and implemented off-site remote education services. However, the Education Ministry did not coordinate these programs, and no minimum standards or spending were required. As a result, each state acted autonomously, leading to a wide array of remote education programs with distinct characteristics concerning the use of technology, the means of transmission, and the financing to sustain the new infrastructure. To determine whether cuts in education funding significantly affected the quality of off-site educational provision, we will test the following hypothesis:

- HYPOTHESIS 1B. States that enacted higher education budget cuts are more likely to have implemented and maintained remote education plans of poorer quality.

Historically, Brazil's federal government, states, and municipalities struggled to collaborate effectively on shared responsibilities in education. Furthermore, the advances were influenced by the varying managerial, technical, and financial capabilities of federated entities (Abrucio et al. 2020; Ranieri 2024). For

this reason, we also seek to examine the potential positive or negative impacts of preexisting state capacity on the provision of public education. Federations with higher intergovernmental cooperation presented better policy responses to COVID-19 (Greer et al. 2023). In Brazil, federal coordination of education expenditures could have been implemented considering local needs and alleviating the burden of implementing emergency educational policy, especially for low-capacity states. However, if the lack of coordination left poorer states unassisted and unprepared for implementation, the remote plans could exacerbate existing inequalities between states. We anticipate the latter and propose hypotheses 2A, 2B, and 2C, that previous state capacity is correlated with remote education plans, divided into the following three hypotheses:

- HYPOTHESIS 2A. Lower preexisting education outcomes are correlated with poorer implementation and maintenance of remote education plans
- HYPOTHESIS 2B. Lower levels of internet access are correlated with poorer implementation and maintenance of remote education plans.
- HYPOTHESIS 2C. Lower state wealth is correlated with poorer implementation and maintenance of remote education plans.

Hypotheses 2A, 2B, and 2C are based on the premise that states with higher capacities, wealth, and prior investments in education and internet infrastructure will be better equipped to implement effective remote education programs. Conversely, states with lower capacities, wealth, and investments in these areas will likely struggle to provide adequate remote education and may enact poorer-quality remote education programs. For instance, on the one hand, states with greater capacities and wealth can invest in robust remote education infrastructure, including technology and teacher training, resulting in more effective remote education programs. On the other hand, states with lower capacities and wealth will face challenges in implementing adequate remote education, leading to a decline in the quality of education provision.

Data and Method

The main dependent variable in this study is the quality and extent of remote education programs, which we measure using a novel indicator called the Remote Education Policy Index (REP Index) (Barberia, Cantarelli, and Schmalz 2021).¹ The REP Index is based on coded documents published by official government sources. Data collection primarily relied on state-level executive decrees

¹ The data and replication materials for the statistical analyses are available on the authors' Harvard Dataverse.

published by state governments and remote education program information published by education departments. In many cases, education departments of local governments published limited information on these programs on their websites. Therefore, we relied on secondary sources like newspapers and government press releases. When official information was unavailable, we presumed the local government did not implement a remote education program.

Table 1 presents the four policies that comprise the REP Index. First, we coded policies to capture the types of media channels used to broadcast remote education, including videos or educational content to students, such as radio, television, and the internet. Second, we also sought to determine which resources were provided to students and teachers to access classes and content, such as cell phones, tablets, textbooks, or any subsidies for internet access. Third, we recorded the policies adopted to ensure student supervision and attendance and whether teachers, institutions, or both were responsible for this supervision. Finally, we recorded which educational levels were included in the remote education program, such as kindergarten, elementary, and high school. Each component was scored 0 (no policy), 0.5 (policy partially offered), or 1 (policy offered).

The three indicators (*B1*, *A1*, and *S1*) measure policies directed at the broadcast, access, and supervision of students in remote education programs; *C1* is a measure of the overall coverage of the population. An aggregate index was calculated for each unit-day using the following formula:

$$\text{REP Index} = \frac{B1 + A1 + S1}{3} \times C1.$$

Thus, the logic of the index is to weigh the first three indicators equally, and this average is then multiplied by the number of education levels covered to

TABLE 1
THE INDICATORS EMPLOYED TO MEASURE THE REMOTE EDUCATION POLICY (REP) INDEX

Indicator	Name	Components	Score
<i>B1</i>	Broadcast technologies	0 = No broadcast offered +1/3 = Internet broadcast +1/3 = Television broadcast +1/3 = Radio broadcast	0–1
<i>A1</i>	Means of access	0 = No access offered +1/3 = Printed materials +1/3 = Gadgets distributed +1/3 = Internet subsidy	0–1
<i>S1</i>	Supervising of students	0 = No supervision planned +1/2 = Teacher supervision +1/2 = Education department supervision	0–1
<i>C1</i>	Coverage	0 = No coverage offered +1/3 = Kindergarten coverage +1/3 = Elementary school coverage +1/3 = High school coverage	0–1
REP Index	REP Index	$(B1 + A1 + S1) \times C1$	0–10 (rescaled)

indicate the comprehensiveness of each plan. The REP Index is then rescaled to a range of 0–10, with a score for each unit-day in states. The lowest score is 0 (indicating no program) and the highest possible score is 10.

We define 2020 as the “year of implementation,” contrasting it with 2021, which we term the “year of maintenance.” This distinction arises because, in 2020, states had to adapt quickly, utilizing existing budgets and resources. Whereas in 2021, states had the opportunity to build on the lessons learned in 2020, enabling them to maintain and expand their programs more effectively.

The school year in Brazil starts in February, and before 2 months of on-site schooling, all states implemented broad school closure policies. The subsequent implementation of remote education programs was slow and heterogeneous, taking 41 days on average and more than 100 days for some states. We attribute daily values to the REP Index beginning from the date of the first school closures. For 2020, states that delayed the implementation of remote education programs were assigned a score of zero for the period before their programs’ initiation. Consequently, the 2020 average REP Index reflects not only the average quality of the plans implemented during the school year but also strongly rewards states that acted promptly and penalizes those that failed to introduce off-site programs. In 2021, remote education remained important as a primary educational service, and accordingly, the REP Index measures how states maintained and further improved 2020 programs in the context of school systems that opted to continue to offer off-site schooling,

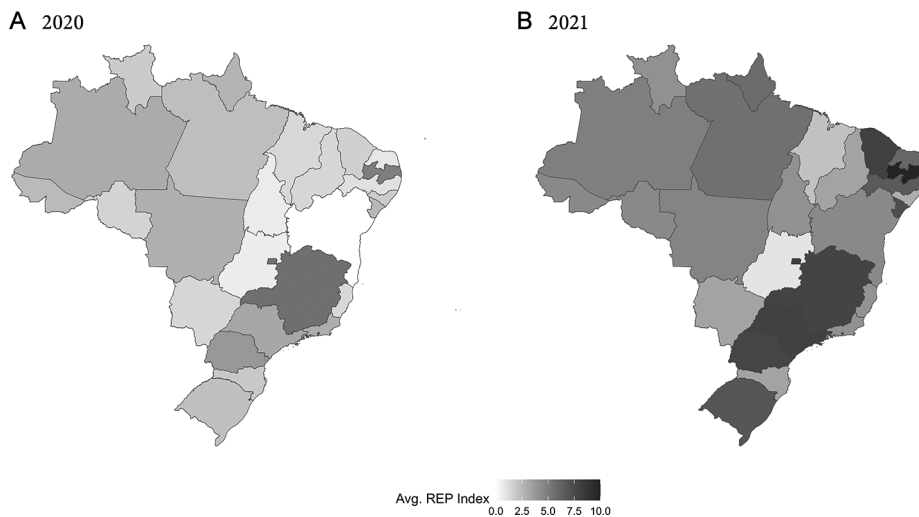


FIG. 1.—Average Remote Education Program (REP) Index for each state of Brazil in A, 2020 (implementation phase) and B, 2021 (maintenance phase). SOURCE.—COVID-19 Government Response Tracker for the Brazilian Federation (CGBRT-FED).

including as hybrid education was introduced once teachers and staff began to receive COVID-19 vaccines.

Figure 1 shows the mean of the REP Index in Brazilian states in 2020 and 2021. On average, the REP score was 2.285 for states in 2020 and 5.310 in 2021. The first year of the pandemic is the implementation phase. In this phase, remote plans were insufficient in most states and were deficient mainly in terms of the types of access to education offered to students, that is, means and alternatives for students to access the available content. Some units provided nearly no plan, with states such as Bahia, Goiás, Tocantins, and Rio Grande do Norte scoring values less than one on average. Only two states (Distrito Federal and Minas Gerais) scored higher than 5 on a scale of 0–10. Programs that received higher scores did so because they offered multiple means of access, but most states did not provide these conditions.

In 2021, which we consider as a year of a new maintenance phase, there was a substantial improvement in most plans, causing states to receive higher values in the REP Index. Most states adopted some form of hybrid return to on-site schooling and improved remote education programs. The most significant improvements were mainly in the access dimension as states diversified how off-site education was delivered to students to cover a more diverse outreach strategy. For this year, only Goiás scored less than 1, followed by Maranhão (with 2.17) and Alagoas (2.99). São Paulo, Ceará, Distrito Federal, and Paraíba scored higher than 8, with Paraíba achieving the highest possible result, 10. As figure 1 confirms, the improvements in the remote education program score were uneven across the states of the federation.

The explanatory variables used in this study are the net change in education expenditures in 2020 relative to 2019, broadband internet access, per capita gross domestic product (GDP), and an indicator of the quality of education. To evaluate the effect of budget cuts, we use the reduction in education expenditures that exceed the reduction in tax revenue in state i in 2020 compared with 2019 (with both measures in constant 2018 R\$). Ideally, the net reduction should have been zero; therefore, a higher net reduction suggests we expect a greater decline in education provision.

To capture preexisting internet access, we use the percentage of households with access to broadband internet in 2019, using data from the Pesquisa Nacional por Amostra de Domicílios Contínua (Instituto Brasileiro de Geografia e Estatística [IBGE] 2019). We use the value of the Basic Education Development Index (IDEB) in state i in 2019 (Instituto Nacional de Estudos e Pesquisa [INEP] 2023) to capture the overall quality of the state educational system. To evaluate income levels and how they might affect the quality of remote education, we include per capita GDP in state i in 2018 (in R\$ thousands).

As controls, we use the mean value of the Social Distancing Policy Stringency Index in state i in 2020 (Barberia, de Faria Oliveira et al. 2021). This measure captures the stringency of nonpharmaceutical interventions against

the pandemic and is a crucial control because better plans may be correlated with the intensity of school closures and other restrictive policies. We also control for state debt negotiation during the pandemic with the Debt Gain variable. Complementary Law 173/20 suspended debts between states and the federal government during 2020, allowing for considerable budget relief in some cases. This regulation was unequally distributed, benefiting only a few states, with substantial gains concentrated mainly in São Paulo and Goiás states, with indirect gains of R\$4.77 billion and R\$1.14 billion, respectively.

To test hypothesis 1A, we present a descriptive analysis of the net losses in education expenditures in 2020. Next, we employ multivariate regression models where the dependent variable is the mean REP score in the year of implementation (2020) and maintenance of remote education (2021) to test hypotheses 1B, 2A, 2B, and 2C. Because there are only 26 observations for each year, we chose to estimate a parsimonious model with a limited number of control variables.

Results

We first describe the main findings regarding net expenditure reductions in education (hypothesis 1A). Figure 2 summarizes the net reduction in education expenditures for each state. Our calculations show that no state maintained or increased education spending in 2020 compared with 2019. This is exemplified by significant decreases in net education expenditure,

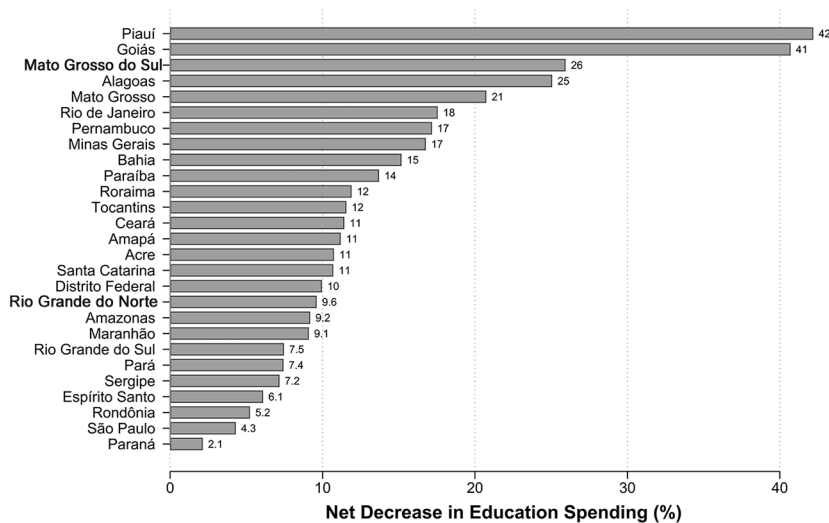


FIG. 2.—Percentage decrease in education spending by state in 2020 (net of revenue loss considering 2019 levels). SOURCE.—Brazilian Public Sector Accounting and Fiscal Information System (Siconfi), <https://siconfi.tesouro.gov.br/siconfi/index.jsf>.

ranging from a decrease of 2.1 percent in Paraná to more than a 40 percent reduction in education spending in Piauí and Goiás in 2020 relative to 2019.

Despite the additional federal resources, states cut education spending at varying levels. As a result, average state net expenditures on education in 2020 and 2021 were lower than 2019 values, even with the supplementation that ensured similar revenues. In our view, the savings from the closure of on-site schooling (represented, e.g., by the reduction in water and energy costs) do not justify the decrease in education expenditures for two reasons. First, states had to undertake additional spending to develop remote education programs and their subsequent implementation. Second, states also had to allocate resources for infrastructure expenses to ensure improved on-site schooling activities in 2021 to ensure safer school environments. These data suggest that the cuts in education were not driven by fiscal considerations but rather by political priorities such as investing in health or pursuing budgetary austerity.

To test hypothesis 1B, namely that, all else being equal, states with larger education budget cuts are more likely to have implemented poorer remote education plans, figure 3 presents the 95 percent confidence interval for the coefficient on the net expenditure education reduction variable for the four regression models that were estimated (see table 2 for full results). We report the coefficient estimates using “Net Education Expenditure Decrease” for the

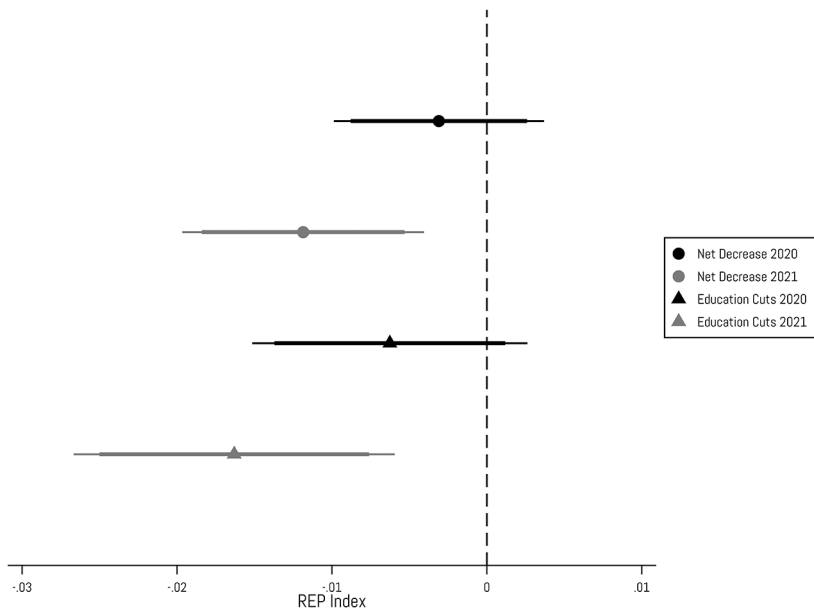


FIG. 3.—Effect of education spending cuts on the Remote Education Program (REP) Index in 2020 and 2021.

TABLE 2
THE EFFECT OF EDUCATION SPENDING AND PREEXISTING INEQUALITIES ON THE
REMOTE EDUCATION POLICY (REP) INDEX

	(1)	(2)	(3)	(4)
Variables	REP Index (2020)	REP Index (2021)	REP Index (2020)	REP Index (2021)
Decrease in education expenditures (percent decrease in 2020 net of revenue loss considering 2019 levels)	-.003 (.003)	-.012** (.004)		
Nominal education expenditure reduction (2020 in R\$)			-.006 (.005)	-.016** (.005)
GDP per capita (2019)	.005+ (.003)	0 (.003)	.005 (.003)	-.001 (.003)
Social distancing policy stringency (2020)	0 (.006)	0 (.006)	.001 (.005)	.003 (.006)
IDEB (2019)	.001 (.074)	-.016 (.085)	.023 (.075)	.029 (.088)
Debt renegotiation gains (2020)	0 (.001)	0 (.001)	-.001 (.001)	0 (.001)
Broadband internet (% of house- holds with access) (2019)	-.001 (.003)	.008* (.004)	0 (.003)	.009* (.004)
Constant	.220 (.447)	.170 (.514)	.076 (.448)	-.207 (.523)
Observations	27	27	27	27
R ²	.268	.414	.305	.426

NOTE.—Standard errors are in parentheses. GDP = gross domestic product; IDEB = Basic Education Development Index.

+ $p < .10$.

* $p < .05$.

** $p < .01$.

REP Index for 2020 and 2021. We also estimated models with the cuts in education in nominal amounts by calculating the difference between 2020 and 2019 spending, regardless of eventual reductions in tax revenue. These results are also reported in figure 3.

As figure 3 confirms, the evidence suggests a negative and statistically significant correlation between states that reduced net education expenditures in 2020 and lower-quality remote education programs in 2021, all else equal. In the second year of the pandemic, the size of budget cuts negatively affected the implementation of plans, considering a 95 percent confidence interval. As a result, states that made more extensive cuts in 2020 were more likely to lag in developing and improving remote education programs as the pandemic's toll progressed. Therefore, we find evidence supporting the hypothesis that states which cut less from education in 2020 were more likely to implement and maintain better plans in 2021. Because we use the lag of expenditures, our findings are compelling in showing that net expenditure cuts in 2020 affect future remote education program quality.

Our results contradict research that has pointed to positive outcomes resulting from the decentralization of COVID-19 policies (Bennouna et al. 2021; Angelici et al. 2023). On the other hand, our findings are aligned with studies that reported on the negative effects of decentralization on public policy during the pandemic around the world (Benton 2020; Bowling et al. 2020; Greer et al. 2023) and in Brazil (Castro et al. 2021). In the area of education, we show that Bolsonaro's laissez-faire strategy reduced state education expenditures and education.

Hypotheses 2A, 2B, and 2C aim to verify whether preexisting state capacity, measured by education provision, internet coverage, and state wealth, is correlated with poorer quality in remote education plans, all else being equal. The effects of these coefficients on the overall REP Index are reported in table 2, and the results of the six models estimated to assess the effects of preexisting inequalities on specific components of the quality of remote education programs are presented in table 3. We employ three subcategories of the REP Index, namely means of access, gadgets, and internet subsidy, as the dependent

TABLE 3
THE EFFECT OF PREEXISTING INEQUALITIES ON INTERNET ACCESS, DISTRIBUTION OF GADGETS,
AND INTERNET SUBSIDIES IN 2020 AND 2021

Variables	(1) Means of Access (2020)	(2) Means of Access (2021)	(3) Gadgets (2020)	(4) Gadgets (2021)	(5) Internet Subsidy (2020)	(6) Internet Subsidy (2021)
Broadband internet access (2019)	-.001 (.003)	.007 (.005)	-.001 (.002)	.006 (.008)	.003 (.005)	.015* (.007)
IDEB (2019)	.024 (.079)	.052 (.106)	-.011 (.039)	.067 (.174)	.221* (.104)	.269 (.162)
GDP per capita (2019)	.006* (.003)	.002 (.004)	.002 (.001)	-.002 (.006)	.004 (.004)	-.002 (.006)
Debt renegotiation gains (2020)	-.001+ (.001)	0 (.001)	0 (0)	.002 (.002)	-.003** (.001)	-.002 (.002)
Decrease in education expenditures (percent decrease in 2020 net of revenue loss considering 2019 levels)	-.006 (.004)	-.007 (.005)	-.001 (.002)	.002 (.008)	-.008 (.005)	-.0141+ (.008)
Social distancing policy stringency (2020)	.001 (.006)	.007 (.008)	-.001 (.003)	-.0104 (.013)	.008 (.008)	.031* (.012)
Constant	.153 (.473)	-.372 (.640)	.119 (.234)	-.042 (1.047)	-1.210+ (.628)	-2.624* (.974)
Observations	27	27	27	27	27	27
R ²	.448	.280	.190	.164	.607	.516

NOTE.—Standard errors are in parentheses. GDP = gross domestic product; IDEB = Basic Education Development Index.

+ $p < .10$.

* $p < .05$.

** $p < .01$.

variables to test whether preexisting capacities are correlated with the adoption of specific components of remote education that ensure programs reach a broader array of students in 2020 and 2021.

Figures 4, 5, and 6 display the 95 percent confidence interval for the coefficient of each explanatory variable, measuring how preexisting inequalities affect the overall or a specific subcomponent of the REP Index in 2020 and 2021. Figure 4 examines the association between state GDP per capita in 2019 and how a state performed in implementing (in 2020) and maintaining (in 2021) a remote education program, as well as how it invested in improving the means of access, distribution of technology gadgets, and subsidies for internet access. States with higher preexisting GDP introduced higher-quality remote internet programs and were more successful in providing access to the internet for their students in 2020. These results confirm that state wealth was correlated with more significant progress in ensuring wider access to remote education in 2020, all else being equal. However, this pattern was not verified in 2021.

Figure 5 examines the extent to which preexisting average state IDEB scores are associated with the development of each subcomponent of remote education. For the internet subsidy variable, which captured the distribution of SIM cards with 3G/4G internet to students or offering applications with free internet access, states with higher scores on IDEB in 2019 proposed

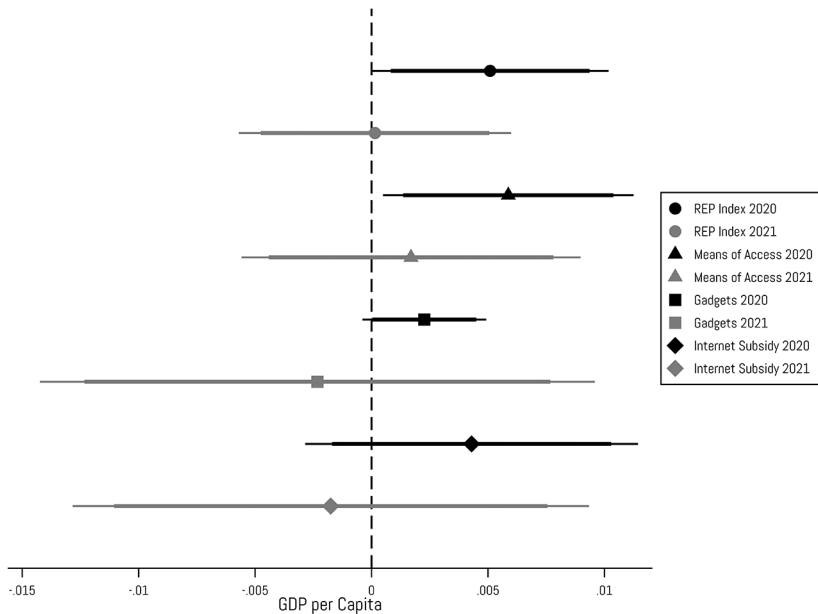


FIG. 4.—Effect of gross domestic product (GDP) per capita on the overall Remote Education Program (REP) Index and its subcomponents.

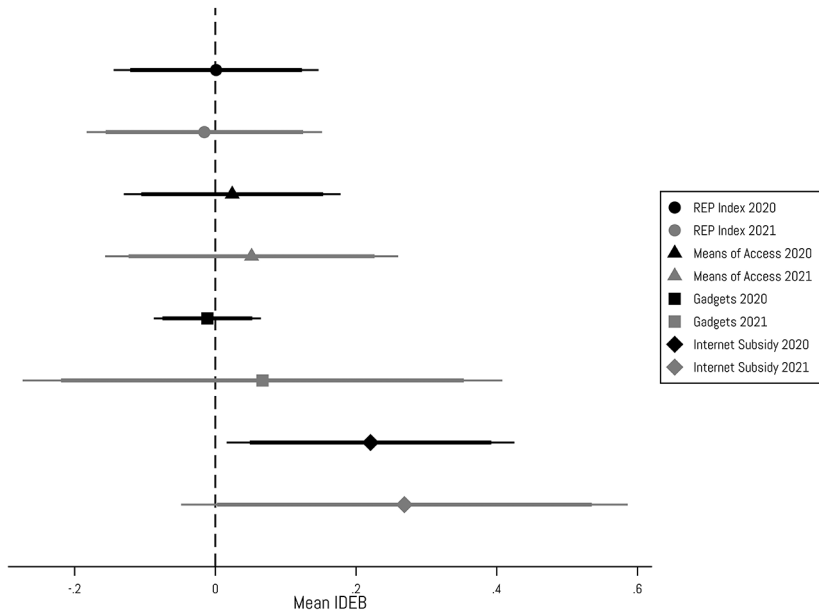


FIG. 5.—Effect of school quality on the overall Remote Education Program (REP) Index and its subcomponents. IDEB = Basic Education Development Index.

better plans to ensure internet access for students in 2020. However, the mean IDEB score in 2019 did not have a statistically significant effect on the likelihood that states would provide internet subsidies in 2021.

Last, figure 6 focuses on the association between preexisting broadband connectivity in 2019 and the quality of remote education programs in 2020 and 2021. Wealthier states, with a larger percentage of the population that had broadband internet access, maintained higher-quality remote education plans for 2021. However, internet coverage did not determine which states introduced higher-quality programs in 2020. States were also more likely to grant internet access subsidies in 2021 if the population had better access to the internet.

In summary, our study provides partial support for our hypotheses. We find that allocating sufficient educational resources during the COVID-19 pandemic was an important determinant of how much states sustained their commitment to providing off-site education. Our results demonstrate that considering the context in which there was a marked decentralization of education management to subnational units in Brazil, states that reduced education expenditures in 2020 had lower-quality remote education programs in 2021. States with higher income, higher educational levels, and internet access before the pandemic implemented higher-quality remote programs and were

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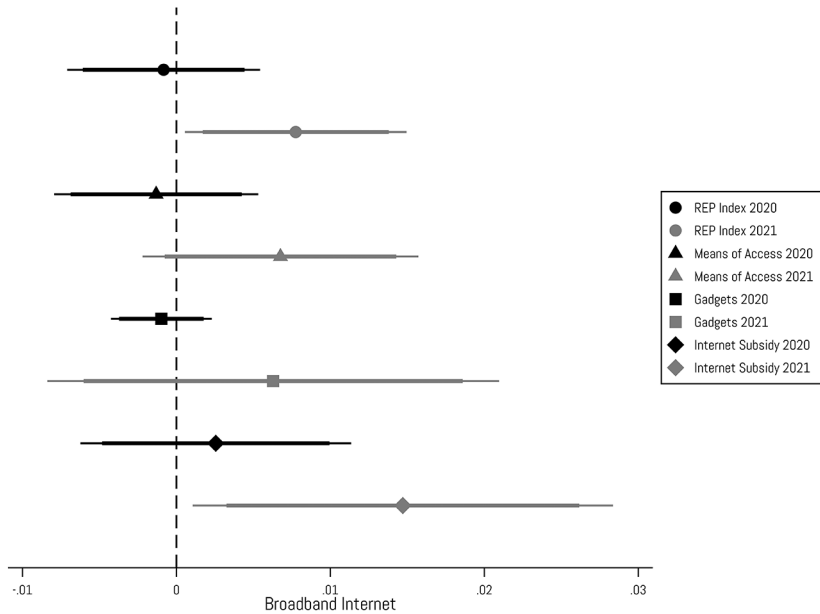


FIG. 6.—Effect of preexisting broadband internet access on the overall Remote Education Program (REP) Index and its subcomponents.

more concerned with subsidizing internet access for their students in 2020. Our findings also show that the federal government’s laissez-faire education response to COVID-19 increased interstate inequalities in education provision, and higher-capacity states provided better plans for remote education.

Conclusion

The global pandemic demanded alternative methods of providing education, presenting a new challenge for federative arrangements of policy provision (Li et al. 2023; Kim et al. 2021). Drawing on the Brazilian case, our study demonstrates that, in the absence of federal coordination, decentralized management of education policy during the pandemic led to significant disparities in the quality and comprehensiveness of remote education programs across states.

We show that the federal government’s loosening of mandatory education expenditure levels resulted in a net decrease in the allocation of resources to education. Most states reduced their education budgets in 2020 and 2021, and education departments enacted cuts of up to 42 percent in 2020 compared with the previous year. In other words, federal aid that guaranteed state revenues would be maintained at 2019 levels in 2020 was insufficient to ensure

that states allocated the same level of resources for education. We find a strong correlation between cuts in education budgets and the implementation and maintenance of remote education programs, contributing to research on the association between budget cuts and educational outcomes.

This finding aligns with previous scholarship that highlights the risks of excessive decentralization during emergencies (Benton 2020; Bowling et al. 2020; Greer et al. 2021). Our analysis builds on the comparative federalism literature (Peterson 1995; Weingast 1995; Treisman 2007), showing that without effective central mechanisms for coordination and redistribution, subnational capacity gaps tend to widen during crises. In line with studies by Carnoy et al. (2017) and Hinchliffe (1989), we show that fiscal and administrative decentralization, when not counterbalanced by federal guidance and financial equalization, can exacerbate educational inequalities.

Once the net decrease in education spending is considered, the heterogeneity of remote education programs across states correlated with previous state characteristics. Therefore, states with lower prepandemic economic and educational levels implemented lower-quality and less comprehensive remote education programs, particularly concerning spending on internet access. Finally, efforts to measure policy, such as the REP Index, are instrumental in providing real-time indicators of the quality and comprehensiveness of education programs.

Beyond the Brazilian case, our findings contribute to broader comparative debates on the role of federal governance in crisis management and response. They suggest that federations with fragmented educational systems, and without mechanisms for horizontal or vertical coordination, are particularly vulnerable to deepening inequalities during global emergencies. Comparative research across federal education systems could further explore how constitutional frameworks, intergovernmental relations, and political leadership shape education resilience. Understanding these dynamics will be crucial for designing more equitable and coordinated responses to future global crises in federal contexts.

Positionality Statement

We position ourselves as scholars committed to reducing inequalities in education. Our research seeks to highlight how public policies affect fair access to schooling. During the COVID-19 pandemic, we joined the Solidarity Research Network, combining timely academic analysis with a broader defense of equity in Brazil's education system. We acknowledge the University of São Paulo as a leading institution in Latin America, while recognizing that higher education in Brazil remains deeply unequal. Despite progress in quotas and inclusion, barriers persist for Indigenous, Afro-Brazilian, and low-income students. We reaffirm our responsibility to confront these inequities through research and teaching.

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